3 Discrete Random Variables and Probability Distributions

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3.1 Random Variables

Random Variables

In any experiment, there are numerous characteristics that can be observed or measured.

For example, in a study of commuting patterns in a metropolitan area, each individual in a sample might be asked about commuting distance and the number of people commuting in the same vehicle, but not about IQ, income, family size, and other such characteristics.

In general, each outcome of an experiment can be associated with a number by specifying a rule of association (e.g., the number among the sample of ten components that fail to last 1000 hours or the total weight of baggage for a sample of 25 airline passengers).

Random Variables

Such a rule of association is called a **random variable**—a variable because different numerical values are possible and random because the observed value depends on which of the possible experimental outcomes results

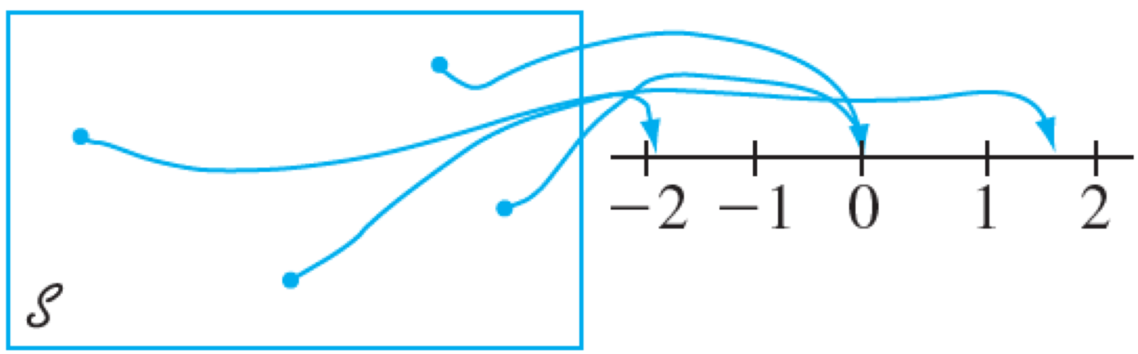

Figure 3.1 A random variable

Random Variables

Definition

For a given sample space $\mathcal{S}$ of some experiment, a **random variable (rv)** is any rule that associates a number with each outcome in $\mathcal{S}$.

In mathematical language, a random variable is a function whose domain is the sample space and whose range is the set of real numbers.

Random Variables

Random variables are customarily denoted by uppercase letters, such as X and Y, near the end of our alphabet.

In contrast to our previous use of a lowercase letter, such as x, to denote a variable, we will now use lowercase letters to represent some particular value of the corresponding random variable.

The notation $X(s) = x$:

means that x is the value associated with the outcome s by the rv *X.*

Example 1

When a student calls a university help desk for technical support, he/she will either immediately be able to speak to someone (*S*, for success) or will be placed on hold (*F*, for failure).

With $\mathcal{S} = \{S, F\}$, define an rv X by

$$X(S) = 1 \qquad X(F) = 0$$

The rv X indicates whether (1) or not (0) the student can immediately speak to someone.

Random Variables

Definition

Any random variable whose only possible values are 0 and 1 is called a **Bernoulli random variable.**

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Example 3

Example 2.3 described an experiment in which the number of pumps in use at each of two six-pump gas stations was determined. Define rv's X, Y, and U by

X = the total number of pumps in use at the two stations

Y = the difference between the number of pumps in use at station 1 and the number in use at station 2

U = the maximum of the numbers of pumps in use at the two stations

Example 3

cont'd

If this experiment is performed and $s = (2, 3)$ results, then $X((2, 3)) = 2 + 3 = 5$, so we say that the observed value of X was $x = 5$.

Similarly, the observed value of Y would be $y = 2 – 3 = –1$, and the observed value of U would be $u = \max(2, 3) = 3$.

Example 4

Consider an experiment in which 9-volt batteries are tested until one with an acceptable (S) voltage is obtained. Define an rv X by

X = the number of batteries tested before the experiment terminates

We can write the sample space and the rv *X:*

$$\mathcal{S} = \{S, FS, FFS, \ldots\}.$$

$$X(S) = 1,\ X(FS) = 2,\ X(FFS) = 3, \ldots, X(FFFFFFS) = 7, \text{ and so on.}$$

Two Types of Random Variables

Two Types of Random Variables

Definition

A **discrete** random variable is an rv whose possible values either constitute a finite set or else can be listed in an infinite sequence in which there is a first element, a second element, and so on ("countably" infinite).

A random variable is **continuous** if *both* of the following apply:

1. Its set of possible values consists either of all numbers in a single interval on the number line (possibly infinite in extent, e.g., from $-\infty$ to ∞) or all numbers in a disjoint union of such intervals (e.g., $[0, 10] \cup [20, 30]$).

Two Types of Random Variables

2. No possible value of the variable has positive probability, that is, $P(X = c) = 0$ for any possible value c.

Example 6

All random variables in Examples 3.1 –3.4 are discrete.

As another example, suppose we select married couples at random and do a blood test on each person until we find a husband and wife who both have the same Rh factor.

With X = the number of blood tests to be performed, possible values of X are $D = \{2, 4, 6, 8, \ldots\}$.

Since the possible values have been listed in sequence, X is a discrete rv.